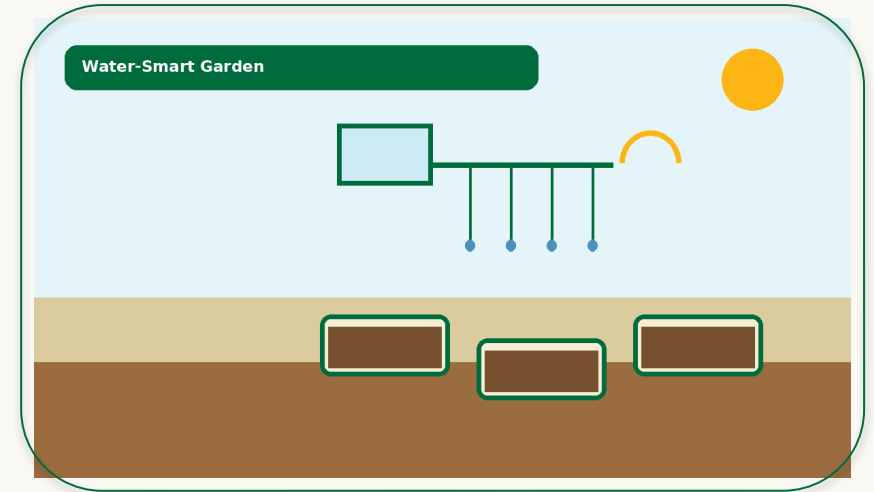


Water-Smart Garden

Agricultural Water Management

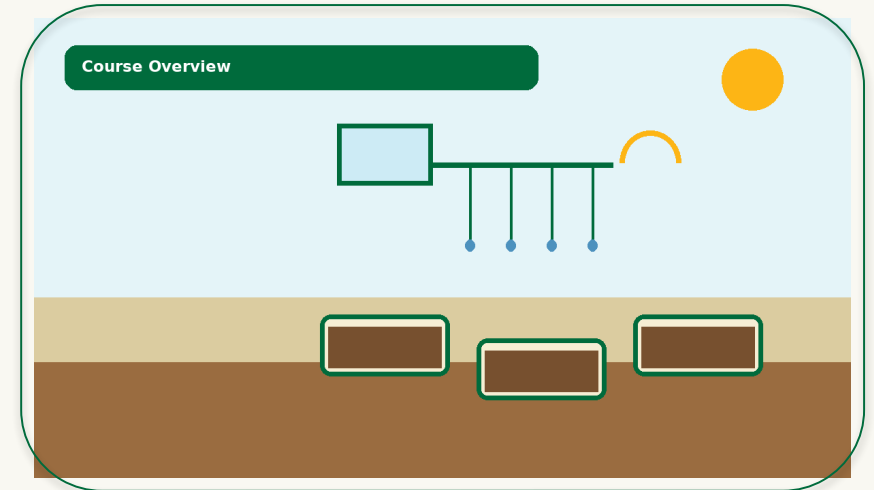
- LIFEWS AgriShine™ GrowMeal™ JSS 1-3 Teacher Training
- Audience: JSS 1-3 teachers, agricultural-science educators and garden facilitators
- Format: Self-paced online course + supervised practical/laboratory orientation
- Recommended duration: 3.5 hours online + 2 hours practical
- JSS learning progression: formulate -> test -> measure -> analyze -> manage



Course Overview

Learning outcomes, materials and teacher readiness

- Learning outcomes: explain the garden/laboratory system; teach all ten technical topics accurately; guide students in measurement, comparison and data collection; connect agricultural science with mathematics and systems reasoning; supervise practical work safely; maintain credible records; interpret evidence without overstating conclusions.
- Core materials: teacher guide, system/plot map, approved tools and irrigation equipment, labels, clipboards, rulers/tapes, graduated containers, meters/sensors where relevant, data sheets, clean sample/harvest containers, PPE/handwashing supplies and first-aid access.
- Completion expectation: complete all modules, demonstrate one safe practical session, and submit a lesson, measurement protocol, experiment or systems investigation appropriate to the course.



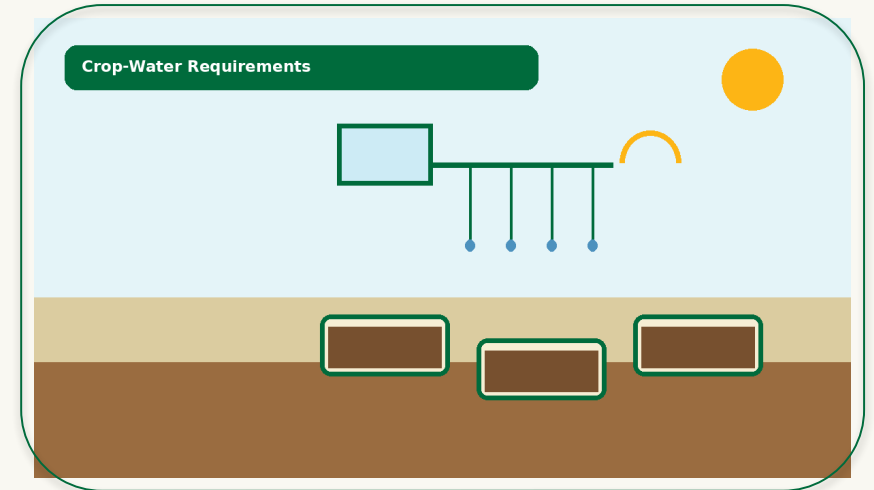
TEACHER PRACTICE

- Training sequence: QUESTION -> PLAN -> IMPLEMENT -> MEASURE -> ANALYZE -> DECIDE.

Crop-Water Requirements

Teacher knowledge, technical practice and student supervision

- Explain crop-water requirement as the water needed to support crop growth under given environmental and management conditions.
- Distinguish crop water need from irrigation amount: rainfall, stored soil water and irrigation can all contribute.
- Discuss how crop type, growth stage, canopy size and weather influence demand.
- Use locally appropriate crop guidance rather than one fixed value for all crops.
- Connect water requirement estimates to irrigation decisions and records.



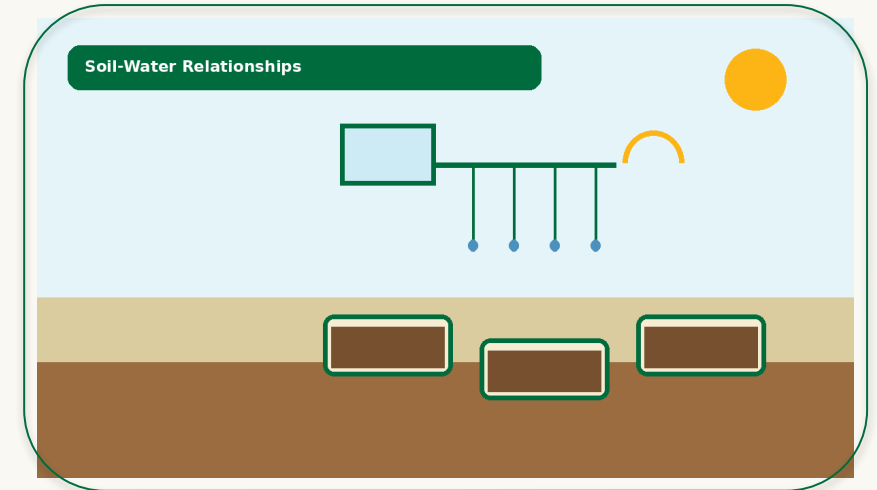
TEACHER PRACTICE

- Teacher task: compare expected water needs at early and later crop stages.

Soil-Water Relationships

Teacher knowledge, technical practice and student supervision

- Explain infiltration, storage, drainage and root uptake as key soil-water processes.
- Relate soil texture and structure to how quickly water enters and how long it may remain available.
- Check moisture at root-zone depth rather than relying only on surface appearance.
- Discuss field capacity and wilting concepts qualitatively at JSS level.
- Avoid interpreting one moisture observation without considering recent irrigation or rainfall.



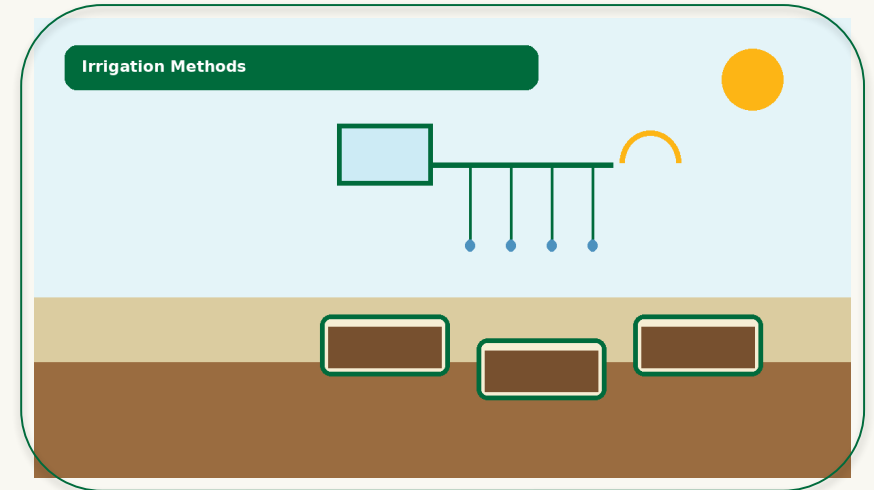
TEACHER PRACTICE

- Prepare a root-zone soil-water observation protocol.

Irrigation Methods

Teacher knowledge, technical practice and student supervision

- Compare hand watering, drip, furrow or other school-relevant methods by delivery pattern, control, labor and potential losses.
- Match the method to crop, soil, slope, water source and garden scale.
- Explain application efficiency conceptually without assigning unsupported percentages.
- Inspect equipment before operation.
- Keep pupils away from pumps, electrical controls and restricted pressure components.



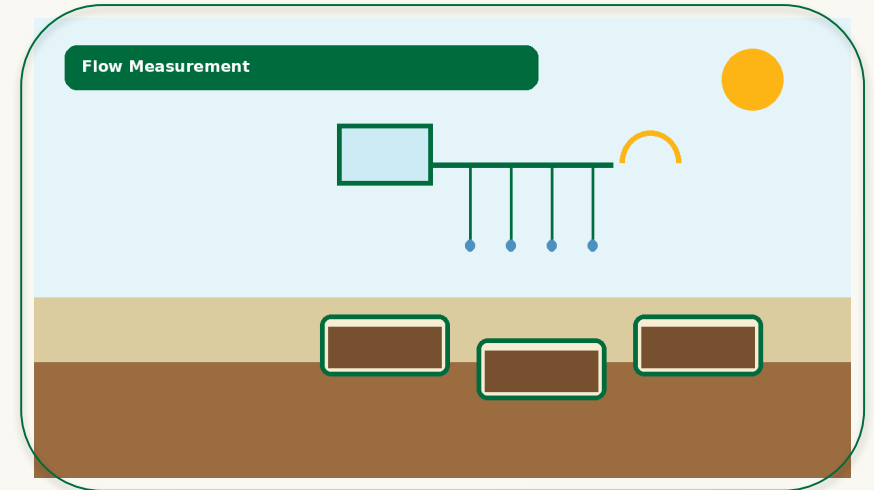
TEACHER PRACTICE

- Create a method-comparison matrix for the school garden.

Flow Measurement

Teacher knowledge, technical practice and student supervision

- Define flow rate as volume of water delivered per unit time.
- Measure flow by timed collection or an installed flow meter.
- Use consistent units such as L/min.
- Repeat measurements to check reliability.
- Use flow data to estimate how much water a zone receives during a known irrigation duration.



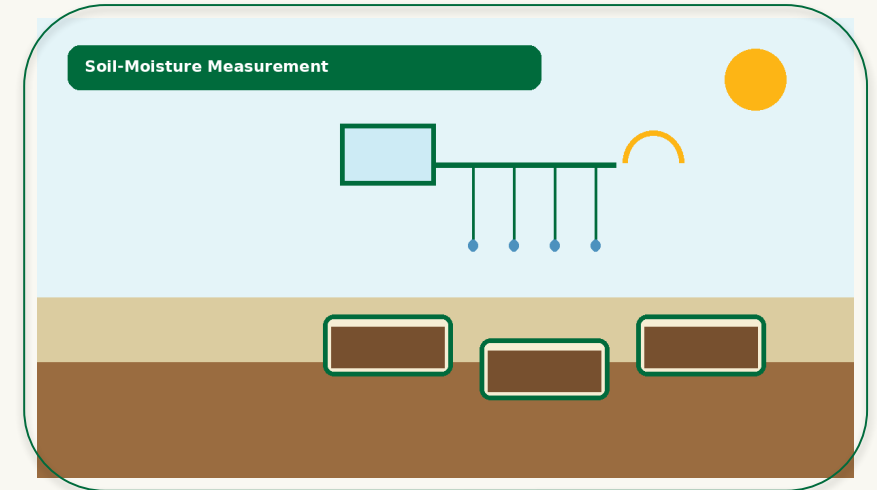
TEACHER PRACTICE

- Measure flow and calculate volume delivered during a 10-minute irrigation.

Soil-Moisture Measurement

Teacher knowledge, technical practice and student supervision

- Compare manual observations with approved soil-moisture sensors where available.
- Standardize measurement depth and location.
- Take readings at multiple points because soil moisture varies spatially.
- Record date, time, irrigation/rainfall context and units or sensor scale.
- Do not treat a single sensor reading as representative of the entire garden without checking.



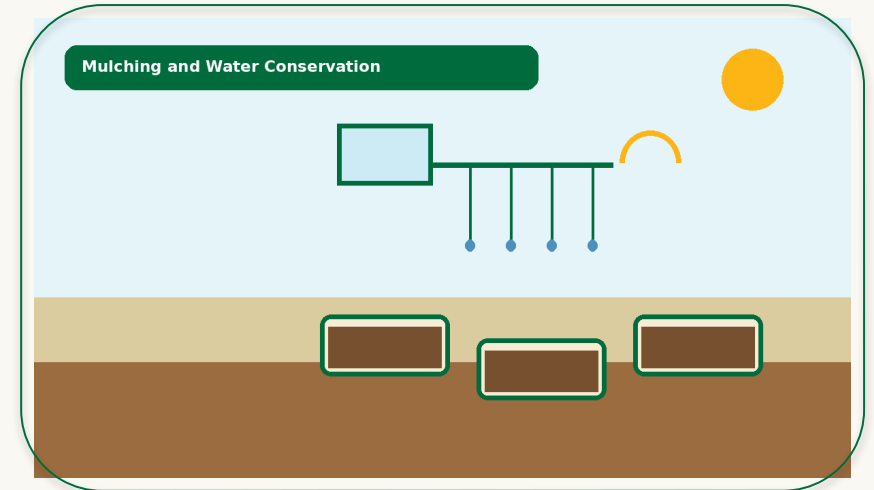
TEACHER PRACTICE

- Design a five-point soil-moisture sampling pattern.

Mulching and Water Conservation

Teacher knowledge, technical practice and student supervision

- Explain how mulch can reduce direct soil-surface evaporation and moderate surface conditions.
- Apply the same approved mulch material and depth when comparing treatments.
- Measure soil moisture and/or irrigation requirement over time.
- Recognize that mulch can also affect temperature, weeds and pest habitat.
- Keep conclusions specific to the observed trial.



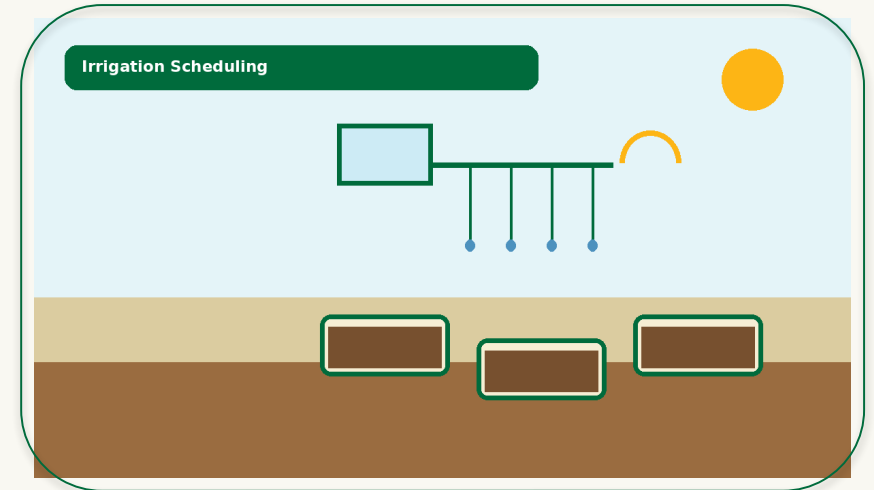
TEACHER PRACTICE

- Set up a replicated mulch versus no-mulch comparison.

Irrigation Scheduling

Teacher knowledge, technical practice and student supervision

- Define scheduling as deciding when to irrigate and how much water to apply.
- Combine crop stage, soil-moisture status, recent rainfall/weather and system capacity.
- Use threshold rules only when they are appropriate to the crop and measurement method.
- Record schedule changes and reasons.
- Avoid deliberately imposing damaging water stress on school production plots.



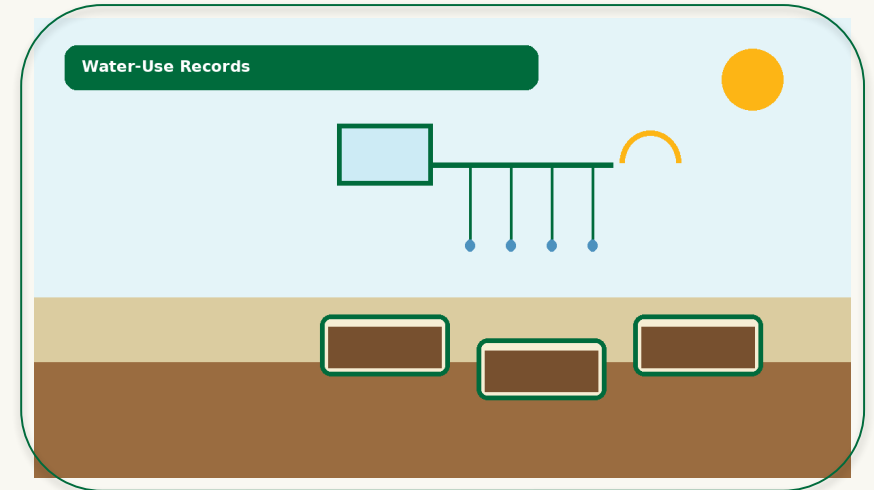
TEACHER PRACTICE

- Develop a one-week evidence-based irrigation schedule.

Water-Use Records

Teacher knowledge, technical practice and student supervision

- Record date, plot/crop, irrigation duration, measured/estimated flow and calculated water volume.
- Include rainfall or other water inputs where measured.
- Use consistent units and distinguish measured from estimated values.
- Check calculations before analysis.
- Use records to identify unusual water-use events such as leaks or repeated irrigation.



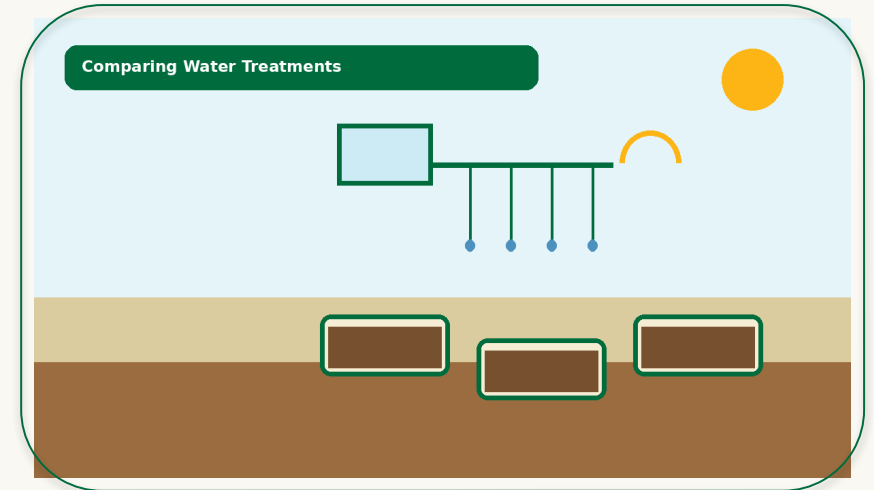
TEACHER PRACTICE

- Create a weekly water-use ledger.

Comparing Water Treatments

Teacher knowledge, technical practice and student supervision

- Define safe treatments such as two approved irrigation schedules, mulch treatments or delivery methods.
- Use controls, replication and consistent crop/plot conditions where feasible.
- Measure water input and relevant crop/soil responses.
- Analyze differences using tables, graphs and simple summary statistics.
- Do not claim superiority from one small trial without considering variability.



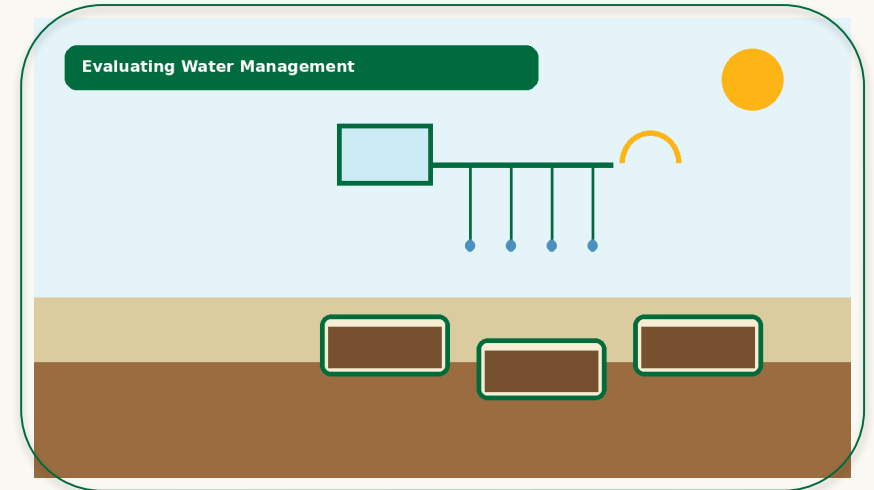
TEACHER PRACTICE

- Design a two-treatment water-management experiment.

Evaluating Water Management

Teacher knowledge, technical practice and student supervision

- Evaluate crop condition/yield, water applied, soil-moisture behavior, labor, system reliability and losses.
- Calculate simple water productivity where appropriate as crop output divided by measured water input, with units stated.
- Discuss trade-offs between saving water and meeting crop needs.
- Use records to recommend practical management changes.
- State limitations and what should be monitored in the next cycle.



TEACHER PRACTICE

- Completion task: prepare an evidence-based water-management evaluation.